

Problem: What is the effect of increasing temperature of water (26.0°C, 30.0°C, 41.5°C, 50.0°C, 55.0°C, and 60.0°C) for dissolving Ba(OH)₂ · 8H₂O, on the solubility constant (K_{sp}) of barium hydroxide determined from hydroxide ion concentration through a titration with hydrochloric acid?

Background Information:

The solubility product constant (K_{sp}) is the equilibrium constant describing the dissolution of a solid substance into an aqueous solution, and is used to describe Dynamic Equilibrium: when the rate of dissolution is equal to the rate of precipitation of solution (Rashe and Peterson). Solubility is the amount of solute that can be dissolved into a solvent at a given temperature (Lu).



Based on the dissociation equation for barium hydroxide, the K_{sp} equation can be found using the concentrations of the products. The concentration of Barium hydroxide is not included as it is a pure solid

$$K_{sp} = [\text{OH}^{-}][\text{Ba}^{2+}]$$

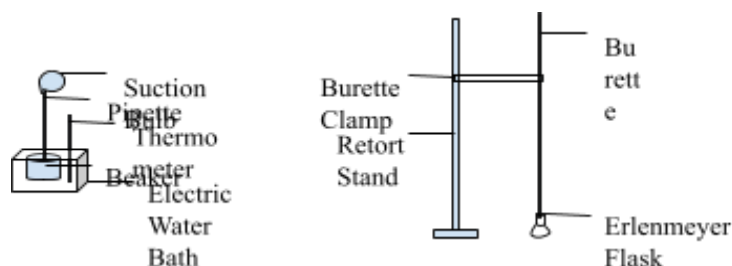
which has fixed “concentration” (Rashe and Peterson).

Barium hydroxide is a strong base, which completely dissociates in water. This is an endothermic process, because the energy required to break the ionic bonds is greater than the energy released. The added heat aids in breaking the ionic lattice that holds the atoms of barium hydroxide together, increasing dissolution.

Hypothesis: If the temperature of the barium hydroxide solution is increased, then the resulting concentration of dissolved compound (and K_{sp}) will also increase because the reaction for dissolution of barium hydroxide is endothermic, with energy as a reactant. Le Chatelier’s principle states that increasing temperature increases the rate of reaction with the higher activation energy. In an endothermic reaction, the forward reaction has a greater activation energy and thus is increased more than the reverse reaction, producing a shift towards the products.

Experimental Design: A Ba(OH)₂ · 8H₂O solution will be placed under six different temperatures in a water bath and tested for solubility by pipetting out 10mL samples of the solution after settling in a water

bath and titrating each one using hydrochloric acid. Three titrations will be conducted for each temperature and averaged.



Manipulated Variable:

Temperature of water used to dissolve a specified amount of barium hydroxide. The temperatures used are 26.0°C, 30.0°C, 41.5°C, 50.0°C, 55.0°C, and 60.0°C. The temperatures chosen range from room temperature to 60.0°C. Higher temperatures lead to significant rates of evaporation. The increments are spaced closer together at both ends (high and low temperatures) to show possible large slope changes with logarithmic scales. This will be controlled using a water bath. The temperature will be measured with a thermometer and recorded to an uncertainty of $\pm 0.1^\circ\text{C}$. The beaker of water will stay in the water bath until three 10.0 mL samples have been pipetted out to ensure the temperature stays constant.

Responding Variable:

The amount of titrant (HCl) needed to fully react the barium hydroxide solution. This will be measured by recording the difference between beginning and end volume measurements on the titrating burette at the moment where the Bromothymol Blue indicator turns permanently yellow. Each volume is measured to an uncertainty of ± 0.1 mL.

Controlled Variables:

Description/Why is it important to maintain?	Controlled/Monitored How?
<i>The type and amount of indicator used in titration:</i> Bromothymol Blue. This is important to maintain to see the same color change between each trial. The color changes at slightly different pH's for different indicators, which would result in a different volume.	Three drops will be added to each erlenmeyer flask for titration using a dropper.

<i>Amount of barium hydroxide octahydrate being dissolved in grams:</i> the amount of solid added can dictate how much is dissolved in the specified time (especially at higher temperatures) because it is more concentrated with more surface area exposed.	Measured into a tray using a scoopula to $8.50\text{g} \pm 0.01\text{g}$ on an electronic mass scale. The same batch of barium hydroxide will be used to ensure the water content is constant.
<i>Amount of solution used for each titration:</i>	Pipetted out using a 10 mL transfer pipette and a silicon suction bulb, ensuring the tip of the pipette does not touch settled precipitate. Measure to the bottom of the meniscus.
<i>Concentration of hydrochloric acid titrant:</i> Different concentrations require different volumes of titrant. Small amounts will add to uncertainty, whereas large amounts are difficult to work with and may have large variation.	0.50 mol/L controlled by using the same batch of hydrochloric acid, which will be standardized beforehand. These will be each titrated thrice and averaged to ensure the accuracy of the standardization.
<i>Amount of water in each beaker:</i>	40mL of water measured using a graduated cylinder to the bottom of the meniscus. Use a distilled water bottle to slowly add water once it nears 40 mL to ensure precision. This ensures enough water for sampling with some leftovers.
<i>Time the dissolved mixture is left to set at the set temperature:</i> leaving the mixture for longer will produce more precipitate which would increase the concentration of ions calculated.	Each mixture will be left in the set water bath temperature for 10 minutes to ensure the solution has time to precipitate. This will be timed using a Samsung A53 timer, and the pipetting will happen immediately.
<i>Titration/Mixing process for the solutions:</i> color change may be present but disappear after mixing, or the final color could differ from a judgement error, which would increase random error in the results.	When the first color change occurs, start swirling the erlenmeyer flask twice for every drop of indicator added. When the color change becomes permanent, stop titrating.

*Since the manipulated variable is temperature, room temperature could skew results if they differ between trials. This should be measured before every titration, but is not a significant source of error because the water bath should maintain a constant temperature despite this.

Procedure: **Wear goggles and use caution when handling chemicals. Immediately rinse if in contact with skin or eyes. Dispose of all waste in a hazardous waste container.*

Standardizing hydrochloric acid titrant:

1. Record room temperature.
2. Set up a retort stand and clamp.
3. Dissolve 7.15g of KHP solid into 70mL of water.
4. Use a 20 mL volumetric pipette and a suction bulb to transfer 20 mL of 0.50M KHP solution into a 50mL erlenmeyer flask.
5. Wash and fill a burette with hydrochloric acid, making sure there is over 20 mL to fully titrate the solution without refilling.
6. Add two drops of phenolphthalein indicator to the flask using a dropper.
7. Lightly swish the flask and place it underneath the burette. Record the initial volume on the burette.
8. Drip the NaOH into the flask, swirling the flask every couple drops when it starts to turn from colourless to pink.
9. Stop titrating when the solution permanently turns pink.
10. Record the final volume on the burette to find the volume of acid used.
11. Repeat this titration with each of the three samples..
12. Repeat steps 2-10 using hydrochloric acid titrant into a fresh sample of the standardized sodium hydroxide. The colour change will go from pink to colorless.

Dissolving solution and Titration

1. Record room temperature.
2. Add water to the water bath to submerge most of a 50mL size beaker without getting water inside. Use a thermometer to set the water bath to the desired temperature and record.
3. Measure out 8.50g of barium hydroxide using an electronic mass scale.
4. Measure out 40mL of water using a graduated cylinder and dissolve the barium hydroxide into the water in the beaker by adding the solid in and thoroughly mixing with a stir stick.
5. Immediately place the beaker into the water bath and roughly stir every two minutes (for eight minutes) then leave alone for 2 minutes.
6. Use a 10mL volumetric transfer pipette to pipette out 10mL of the barium hydroxide solution from the top of the beaker, making sure none of the precipitate at the bottom is pipetted up.
7. Transfer each 10mL sample into a separate 50mL erlenmeyer flask.
8. Repeat Steps 6-7 thrice for three titration trials.
9. Set up a burette stand and clamp.

10. Fill up a burette for acid titration with hydrochloric acid, making sure there is at least 30mL in order to fully titrate the solution at the highest temperature without refilling it.
11. Add two drops of bromothymol blue to each erlenmeyer flask using a dropper.
12. Lightly swish the flask.
13. Place the erlenmeyer flask underneath the burette. Record the initial volume listed on the burette.
14. Slowly drip the hydrochloric acid into the flask, swirling the flask every couple drops when the blue starts to change color.
15. Stop titrating when the solution permanently turns yellow.
16. Record the final volume on the burette to find how much acid was used.
17. Repeat Steps 2-16 for each temperature manipulation.

Raw Data:

Mass of KHP: $7.15\text{g} \pm 0.01\text{g}$

Volume of water: $70.0\text{ mL} \pm 0.1\text{ mL}$

Lab Temperature: $24.5^{\circ}\text{C} \pm 0.5^{\circ}\text{C}$

Qualitative Observations:

1. All solutions were translucent (translucent bright pink to clear)
2. Each titration solution turned a darker shade of pink before suddenly turning clear with the addition of a drop of titrant.
3. As the solutions sat on the lab counter, they retained a clear water-like color.

Table 1: Titration data for standardizing sodium hydroxide using 0.50 mol/L potassium hydrogen phthalate (KHP) and three drops of phenolphthalein indicator.

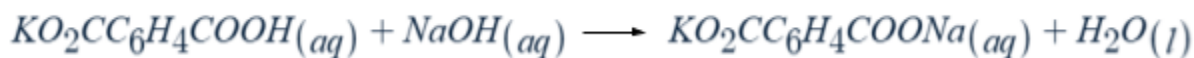
Standardizing NaOH with 0.50mol/L KHP (NaOH Titrant)	Titration 1	Titration 2	Titration 3
Initial Burette Reading ($\pm 0.1\text{ mL}$)	0.9	21.6	22.1
Final Burette Reading ($\pm 0.1\text{ mL}$)	21.6	42.5	43.0
Volume Used ($\pm 0.2\text{ mL}$)	20.7	20.9	20.9

$$\text{Average Volume}(\text{NaOH}) = \frac{20.7\text{mL} + 20.9\text{mL} + 20.9\text{mL}}{3} = 20.8333 \dots \text{mL}$$

Concentration of Potassium Hydrogen Phthalate:

$$C(\text{KHP}) = (7.15\text{g}) \cdot \left(\frac{1\text{mol}}{204.23\text{g}}\right) \cdot \left(\frac{1}{0.070\text{L}}\right) = 0.5001 \dots \text{mol/L KHP}$$

Concentration of Sodium Hydroxide:



$$\begin{array}{l} C = 0.5001 \dots \text{mol/L} \\ V = 0.020 \text{ L} \end{array}$$

$$\begin{array}{l} C = ? \\ V = 0.0208 \dots \text{L} \end{array}$$

$$C(\text{NaOH}) = \left(\frac{0.5001 \dots \text{mol}}{\text{L}} \text{KHP}\right) \cdot (0.020\text{L KHP}) \cdot \left(\frac{1\text{NaOH}}{1\text{KHP}}\right) \cdot \left(\frac{1}{0.0208 \dots \text{L}}\right) = 0.4801 \dots \text{mol/L NaOH}$$

Uncertainty Calculation (Concentration of NaOH):

$$\text{KHP Mass} = \frac{0.01\text{g}}{7.15\text{g}} \cdot 100\% = 0.13986 \dots \%$$

$$\text{KHP Volume} = \frac{0.1\text{mL}}{70.0\text{mL}} \cdot 100\% = 0.14285 \dots \%$$

$$\text{KHP Concentration} = 0.14285 \dots \% + 0.13986 \dots \% = 0.28271 \dots \%$$

$$\text{NaOH Volume} = \frac{0.2\text{mL}}{20.83 \dots \text{mL}} \cdot 100\% = 0.96000 \dots \%$$

$$20\text{mL Pipette} = \frac{0.06\text{mL}}{20.00\text{mL}} \cdot 100\% = 0.3\%$$

$$\text{NaOH Concentration} = 0.28271 \dots \% + 0.3\% + 0.96000 \dots \% = 1.5427 \dots \% = 1.54\%$$

Table 2: Titration data for standardizing hydrochloric acid using 0.48 mol/L sodium hydroxide and three drops of phenolphthalein indicator.

Standardizing HCl with NaOH (HCl Titrant)	Titration 1	Titration 2	Titration 3
Initial Burette Reading (± 0.1 mL)	20.8	21.9	17.0

Final Burette Reading (± 0.1 mL)	39.7	40.9	35.9
Volume Used (± 0.2 mL)	18.9	19.0	18.7

$$\text{Average Volume}(HCl) = \frac{18.9\text{mL} + 19.0\text{mL} + 18.7\text{mL}}{3} = 18.8666 \dots \text{mL}$$

Concentration of Hydrochloric Acid:



$$\begin{array}{ll} C = ? & C = \\ V = 0.0188 \dots L & 0.4801 \dots \text{mol/L} \\ & V = 0.020 L \end{array}$$

$$C(HCl) = \left(\frac{0.4801 \dots \text{mol}}{L} NaOH \right) \cdot (0.020 L) \cdot \left(\frac{1 HCl}{1 NaOH} \right) \cdot (1/0.0188 \dots L) = 0.5089 \dots \text{mol/L } HCl$$

Uncertainty (Hydrochloric Acid Concentration):

The uncertainties for concentration of hydrochloric acid are calculated by adding percent uncertainties as shown above with sodium hydroxide.

$$HCl \text{ Concentration Uncertainty} = 2.9027 \dots \% \text{ or } 0.01477 \dots \frac{\text{mol}}{L}$$

Raw Data:

Lab Temperature: $21.1^\circ\text{C} \pm 0.5^\circ\text{C}$

Qualitative Observations:

1. Bromothymol blue was a deep blue, which turned into a cloudy light blue once added to the barium hydroxide solution.
2. Color changed from cloudy light blue to green to a slightly translucent yellow as the solution settled and titrant was added.
3. First pipette sample was mostly clear, and got cloudy when I got to the third sample because of the limited amount of solution.
4. Large amounts of undissolved barium hydroxide were present in the sampling beaker afterwards.

5. The last two manipulations had larger amounts of barium hydroxide dissolved and was evident because these samples cooled down in the pipette and precipitated making it harder to get out.
6. The water in the water bath and in the beaker started steaming slightly at about 55.0°C

*The volume used in the tables below are the average between the two trials that used less titrant volume and are highlighted in grey in my data table, because the last sample in each manipulation was significantly more cloudy due to solid being pipetted up. The limited amount of original solution was not enough to avoid this.

Table 3: Volume of 0.51 mol/L hydrochloric acid needed to titrate 10.0mL of barium hydroxide solution at $26.0 \pm 0.1^\circ\text{C}$.

	Titration 1	Titration 2	Titration 3
Initial Burette Reading (± 0.1 mL)	0.4	10.6	20.7
Final Burette Reading (± 0.1 mL)	10.6	20.7	31.0
Volume Used (± 0.2 mL)	10.2	10.1	10.3

Table 4: Volume of 0.51 mol/L hydrochloric acid needed to titrate 10.0mL of barium hydroxide solution at $30.0 \pm 0.1^\circ\text{C}$.

	Titration 1	Titration 2	Titration 3
Initial Burette Reading (± 0.1 mL)	0.6	10.1	20.5
Final Burette Reading (± 0.1 mL)	10.1	20.5	31.5
Volume Used (± 0.2 mL)	9.5	10.4	11.0

Table 5: Volume of 0.51 mol/L hydrochloric acid needed to titrate 10.0mL of barium hydroxide solution at $41.5 \pm 0.1^\circ\text{C}$.

	Titration 1	Titration 2	Titration 3
Initial Burette Reading (± 0.1 mL)	0.3	13.7	27.7

Final Burette Reading (± 0.1 mL)	13.7	27.6	41.2
Volume Used (± 0.2 mL)	13.4	13.9	13.5

Table 6: Volume of 0.51 mol/L hydrochloric acid needed to titrate 10.0mL of barium hydroxide solution at $50.0 \pm 0.1^\circ\text{C}$.

	Titration 1	Titration 2	Titration 3
Initial Burette Reading (± 0.1 mL)	0.6	15.3	29.7
Final Burette Reading (± 0.1 mL)	15.3	29.7	47.0
Volume Used (± 0.2 mL)	14.7	14.4	17.3

Table 7: Volume of 0.51 mol/L hydrochloric acid needed to titrate 10.0mL of barium hydroxide solution at $55.0 \pm 0.1^\circ\text{C}$.

	Titration 1	Titration 2	Titration 3
Initial Burette Reading (± 0.1 mL)	3.6	3.6	24.0
Final Burette Reading (± 0.1 mL)	26.7	28.1	45.5
Volume Used (± 0.2 mL)	23.1	24.5	21.5

Table 8: Volume of 0.51 mol/L hydrochloric acid needed to titrate 10.0mL of barium hydroxide solution at $60.0 \pm 0.1^\circ\text{C}$.

	Titration 1	Titration 2	Titration 3
Initial Burette Reading (± 0.1 mL)	0.4	24.8	3.5
Final Burette Reading (± 0.1 mL)	24.8	49.2	28.8
Volume Used (± 0.2 mL)	24.4	24.4	25.3

Sample Repeated Trials Uncertainty Calculation (Temperature: 55.0°C):

$$Average = \frac{(21.5 + 23.1)mL}{2} = 22.3 mL$$

$$|22.3 mL - 21.5 mL| = 0.8 mL$$

$$|22.3 mL - 23.1 mL| = 0.8 mL$$

The average of each data point's difference from the average value is 0.8 mL. This is the repeated trial uncertainty for this temperature manipulation, and will be used because it is larger than the instrument uncertainty of 0.2mL.

Sample Calculation for Ksp (Temperature: 55.0°C):

$$K_{sp} = [Ba^{2+}][OH^-]^2$$

$$K_{sp} = [Ba(OH)_2][2Ba(OH)_2]^2$$

$$K_{sp} = [0.567504598][1.135009196]^2 = 0.731085... = 0.73$$

Ksp uncertainty (Temperature: 55.0°C):

$$[Ba(OH)_2] \% \text{ uncertainty} = HCl \text{ concentration uncertainty} + HCl \text{ volume uncertainty} + Ba(OH)_2 \text{ Sample volume uncertainty}$$

$$[Ba(OH)_2] \% \text{ uncertainty} = 2.9027... \% + \left(\frac{0.8}{22.3} \cdot 100\% \right) + 0.2 \% = 6.69023... \%$$

$$K_{sp} \% \text{ Uncertainty} = 3 \cdot 6.69023... \% = 20.07069... \% = 20.07\%$$

$$K_{sp} \text{ Absolute Uncertainty} = \frac{20.07... \%}{100\%} \cdot 0.7310854... (K_{sp}) = 0.14673... = 0.15$$

Table 9: Processed data formatted to one decimal place. Ksp and Ksp uncertainties are minimal and thus are formatted to two decimal places for precision.

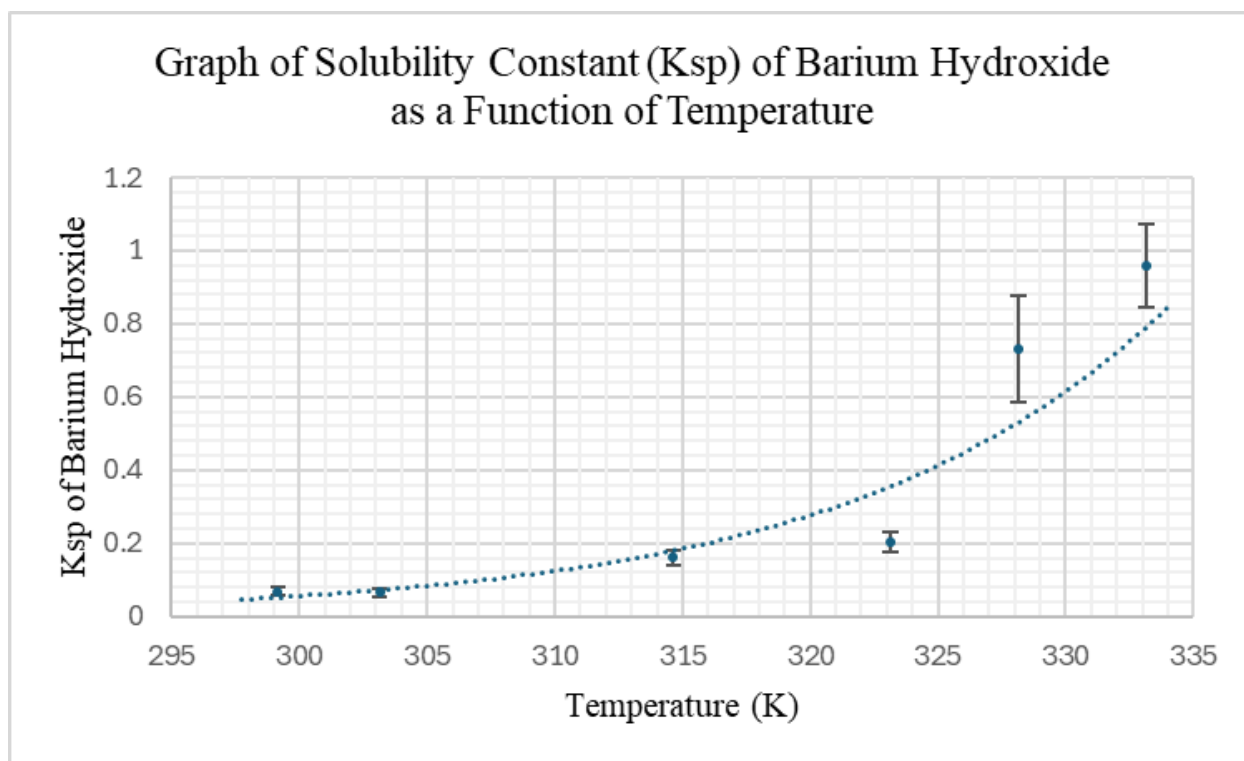
Temperature (±0.1K)	Average Volume of HCl used	HCl Volume Absolute Uncertainty (±)	Barium Hydroxide Concentration	Ksp	Ksp percent uncertainty (%)	Ksp absolute uncertainty
299.2	10.2	0.2*	0.26	0.07	15.22	0.01

303.2	10.0	0.3	0.25	0.06	18.35	0.01
314.7	13.5	0.2*	0.34	0.16	13.77	0.02
323.2	14.6	0.2*	0.37	0.20	13.43	0.03
328.2	22.3	0.8	0.57	0.73	20.07	0.15
333.2	24.4	0.2*	0.62	0.96	11.77	0.11

*An uncertainty of ± 0.2 is used for HCl concentrations that have a repeated trials uncertainty below the 0.2 mL measurement uncertainty.

Graph 1: Solubility Constant (K_{sp}) of Barium Hydroxide as a Function of Temperature (K)

*Horizontal (temperature) uncertainty is minimal as I am confident in my thermometer measurements. Therefore the error is negligible and error bars are not shown.



The effect of increasing temperature on K_{sp} depends on whether the reaction is exothermic or endothermic, which can be deduced using enthalpies. Using Hess' law, enthalpy of a solution, the energy change that occurs during dissolution, can be calculated using the equation below (Chieh).

$$\Delta H_{\text{solution}} = \Delta H_{\text{lattice}} + \Delta H_{\text{hydration}}$$

$$\Delta H_{\text{solution}} = (2 \cdot -460\text{kJ} + -1309\text{kJ}) + (2300\text{kJ}) = +71\text{kJ}$$

Using standard enthalpies from *Wired Chemist* and *Kemu Studio*, the enthalpy of solution of barium hydroxide is calculated to be +71kJ. The positive value tells us that this reaction absorbs energy, thus proving that it is endothermic.

From this graph, we can see an upward curving trend. As temperature increases, the Ksp of Barium Hydroxide increases at a greater rate than the temperature, a relationship modelled by Van't Hoff's equation:

$$\ln K_{\text{eq}} = -\frac{\Delta_r H^\ominus}{RT} + \frac{\Delta_r S^\ominus}{R}$$

This is derived by equating two expressions for Gibbs Free Energy and then dividing both sides by -RT, showing that the equilibrium constant is closely related to enthalpy and entropy calculations. The graph below is curve straightened according to this predicted trend.

$$-RT \ln K_{\text{eq}} = \Delta H^\circ - T \Delta S^\circ$$

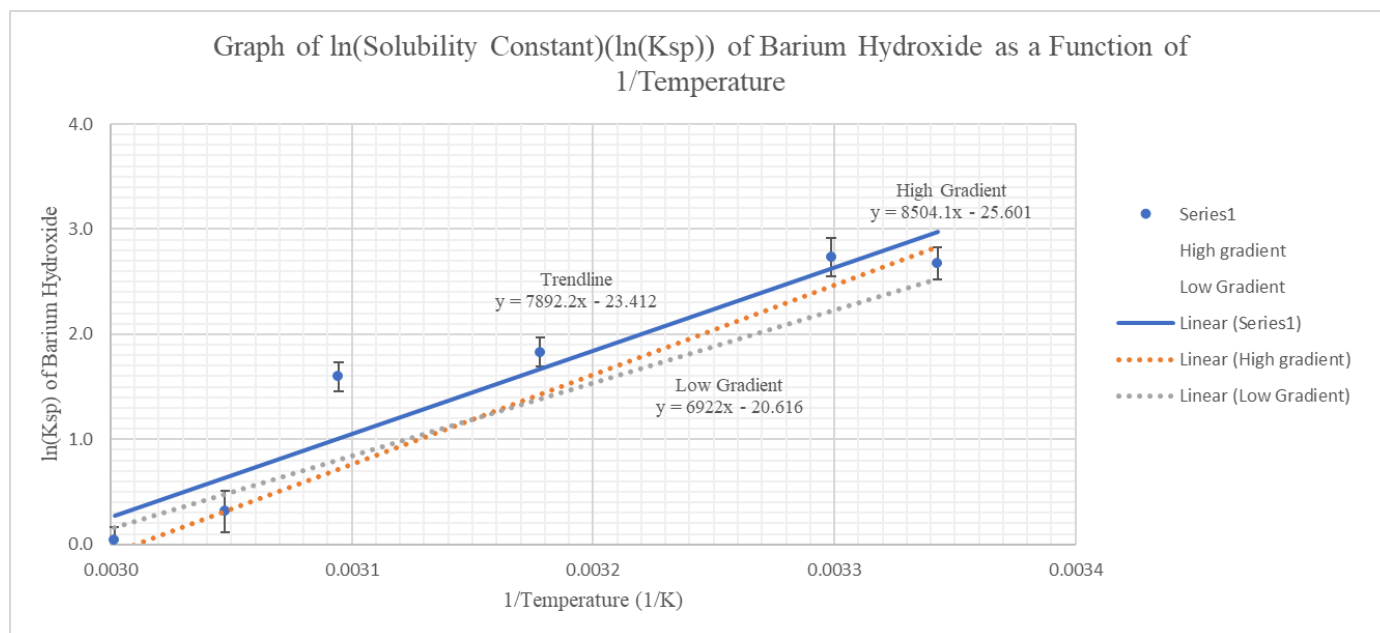
Table 10: Curve Straightening values and Uncertainty for the Effect of Temperature on dissolved Barium Hydroxide ion concentration.

**ln(Ksp) absolute uncertainty is the same as the percent uncertainty for Ksp expressed as a decimal.*

1/Temperature is expressed to two decimal places because of how minimal it is.

1/Temperature (1/K)	-ln(Ksp)	ln(Ksp) absolute uncertainty
0.0033	2.67	0.15
0.0032	2.73	0.18
0.0031	1.83	0.14
0.0031	1.59	0.13
0.0030	0.31	0.20
0.0030	0.04	0.12

Graph 2: -ln of Solubility Constant (-ln[Ksp]) of Barium Hydroxide as a Function of 1/Temperature



(1/K) curve straightened according to Van't Hoff's Equation.

The negative slope corresponds to the change in enthalpy over the gas constant ($\frac{\Delta_r H^\ominus}{R}$) and the y-intercept corresponds to entropy over the gas constant ($\frac{\Delta_r S^\ominus}{R}$), according to Van't Hoff.

Trendline Slope Calculation:

$$\text{Slope} = \frac{\Delta y}{\Delta x} = \frac{2.63 - 0.26}{0.0033 - 0.0030} \text{ mol} \cdot \text{K} = 7900 \text{ mol} \cdot \text{K}$$

The slope of the trend line is 7.9E3 and the slope of the high and low gradients respectively is 8.5E3 and 6.9E3 (mol K), as calculated in the sample above. These are put to only 2 significant digits as this was the lowest number of significant digits used for ln(Ksp), showing a low degree of confidence in the results.

$$|\text{low gradient slope} - \text{trendline slope}| = |6922.0 \text{ mol} \cdot \text{K} - 7892.2 \text{ mol} \cdot \text{K}| = 970.2 \text{ mol} \cdot \text{K}$$

$$|\text{high gradient slope} - \text{trendline slope}| = |8504.1 \text{ mol} \cdot \text{K} - 7892.2 \text{ mol} \cdot \text{K}| = 611.9 \text{ mol} \cdot \text{K}$$

The difference between the low gradient and the slope of the trend line is greater and will therefore be the

one used as the uncertainty. The final slope ($\frac{\Delta_r H^\ominus}{R}$) is $7892.2 \pm 970.2 \text{ mol} \cdot \text{K}$.

Percent Error Calculation:

The percent error can be calculated for my first temperature manipulation using the theoretical K_{sp} value for barium hydroxide octahydrate at 25°C (Room temperature) of 2.55×10^{-4} (CRC Handbook of Chemistry and Physics). My first temperature manipulation happens at 26°C, which is close enough for their values to be compared for accuracy.

$$\begin{aligned} \text{Percent Error} &= \left| \frac{\text{Theoretical} - \text{Experimental}}{\text{Theoretical}} \right| \cdot 100\% \\ \text{Percent Error} &= \left| \frac{2.55 \cdot 10^{-4} - 0.068936 \dots}{2.55 \cdot 10^{-4}} \right| \cdot 100\% \\ &= 2.69 \cdot 10^4 \% \end{aligned}$$

Conclusion:

Changing the temperature of water in the dissolution of barium hydroxide pentahydrate ($\text{Ba}(\text{OH})_2 \cdot 8\text{H}_2\text{O}$) produces an observable change in the amount of solid barium hydroxide that can be dissolved into the solution, thus changing the hydroxide ion concentration dissolved into the solution. According to Van't Hoff's equation, which states that $\ln(K_{sp})$ is proportional to $1/\text{Temperature}$, an increase in temperature increases K_{sp} at equilibrium. The first graph is best fit with an exponential-looking trend line, where a slight increase in temperature produces a significantly larger increase in its solubility constant. After curve straightening by taking the negative natural log of the y-values and inverting temperature on the x-axis, it is evident that as temperature increases, the graph continues upward, showing that *negative* $\ln(K_{sp})$ -for ease of slope calculation and visualization-is increasing (or $\ln(K_{sp})$ decreasing). The general trend supports the theory predicted in the hypothesis and in Van't Hoff's equation, and this method of curve straightening produces a trend line that is most linear, proving that this models the relationship between temperature and K_{sp}. The trendline also supports that of an endothermic reaction, according to Le Chatelier's principle. This confirms the hypothesis that increasing temperatures will lead to a greater K_{sp}.

The slope of the trend line on Graph 2 is 7892.2 molK with an absolute uncertainty of ± 970.2 . This produces a rather large percent uncertainty of around 12%. This slope represents the change in enthalpy (J) over the gas constant, $R = 8.31 \text{ J/mol K}$. Simplifying the units gets mol K as the units for the slope. Since R is constant, the slope multiplied by this gas constant, 8.31J/molK, produces the enthalpy change.

$$\Delta H(\text{enthalpy}) = 7892.2 \text{ mol} \cdot \text{K} \cdot \frac{8.31 \text{ J}}{\text{mol} \cdot \text{K}} = 65584.18 \text{ J} = 65.6 \text{ kJ}$$

This value is less than an 8% error from the theoretical value of enthalpy of solution (71.0kJ) calculated earlier, but this low percent error doesn't improve my experiment as the percent error for K_{sp} was unreasonably large: $2.73 \times 10^4\%$. This is an incredibly huge error that likely happened due to my experimental design, as I will evaluate later. Based on the high uncertainty and even higher percent error, I have an extremely low confidence in my results. This is also because the titration itself produced values that were far different from each other, leading to large variability in my data that the three repeated trials couldn't account for. Though the general trend of the results match up with scientific theory, the issues with my selected methodology take the reliability out of all the data.

In both graphs, there are only two data points that have uncertainty bars passing through the trendline. Although my uncertainties are relatively large, ranging from 12-20%, the majority of my data doesn't coincide with the trendline. This shows that the correlation for the trend is very weak because points are scattered all over the place. With multiple points out of place, it is impossible to identify possible outliers in the data.

Evaluation:

1. The biggest source of error contributing to my enormous percent error is the limited volume of solution present to pipette out of. I needed three 10mL samples from a 40mL solution, but as I got near the bottom of the beaker while pipetting, a significant amount of undissolved solid was pipetted out into the sample. This would make the titrated solution more concentrated than it's supposed to be, leading to a largely inflated K_{sp} . This is a random error because the amount of undissolved solid is different at each temperature, and should be fixed by making a larger amount of solution, and using a thinner beaker. This way, the tip of the pipette could be kept near the top of the solution and not pipette out any solid precipitate or cloudy solution that hadn't settled. The solution should be allowed to settle for about 20s between each sample to allow the disturbed solid at the bottom to settle again.
2. Another huge source of random error is the methodology for mixing the solution in the water bath. Not having the beaker sit for as long of a time, as well as disturbances made the solution more cloudy than expected due to the excessive presence of undissolved ions. As well, the beaker looked significantly more shallow in the water than it actually was, and it was difficult to keep my pipette on the surface of the water. This is a smaller error that would also add to the extra precipitate being in the pipetted sample, leading to a larger K_{sp} . The best improvement for these two errors would be to change my experiment entirely. Warming up a beaker of water to each temperature manipulation and adding barium hydroxide solid to it until it stops dissolving to

measure OH⁻ concentration would be a better way to reduce the errors involved in sampling and titrating.

3. A random error in human judgement would explain the large differences in data obtained. The indicator I used was bromothymol blue, so the color change to yellow was difficult to judge as there are multiple shades of green in between. An improvement for this would be to find a standard color online to compare to, but also to keep each flask after titrating so that I could compare the latter flask's colors to the finished ones. This would ensure that an excess amount of titrant would not be added in order to ensure the color was accurate and consistent with other trials.
4. Evaporation of the solution is also a random error. Although the surface area is the same for each beaker, there is a higher rate of evaporation at higher temperatures due to the increased movement of water molecules in the solution. During my last two temperature manipulations, I could feel that the water was getting quite hot to the touch, and wisps of steam were present. An improvement for this would be to lower my temperature manipulations, and do more manipulations under 50°C rather than going too high.
5. A final systematic error is the change in concentration of the hydrochloric acid titrant over time, as it was standardized and left alone 7 days before my actual trials. This would not skew the trend because the same titrant is still used for each trial, but the concentration may have depleted, meaning more is needed to titrate the sample, thus leading to a higher K_{sp}. However, the HCl was kept enclosed and was only left for 7 days, so the effect of this on the results is likely moderate. This is best improved by doing the standardization of the tyrant right before it will be used.

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